## SIMATS ENGINEERING
## ASSESSMENT TOOLS

**COURSE NAME:** **Theory of Computation** **COURSE CODE:** **CSA1304**

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COURSE OUTCOME COVERED

**CO3: Construct regular expressions and use the pumping lemma and closure properties to analyze regular languages. (BL:3)**

*Assessment Tool Weightage: 40%*

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**QUESTIONS: 5** **Total Marks:100**
**Duration: 1 Hour**

# Experiments

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***Code of Conduct:***

*I, Girija B / 192311044, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student: Girija B***

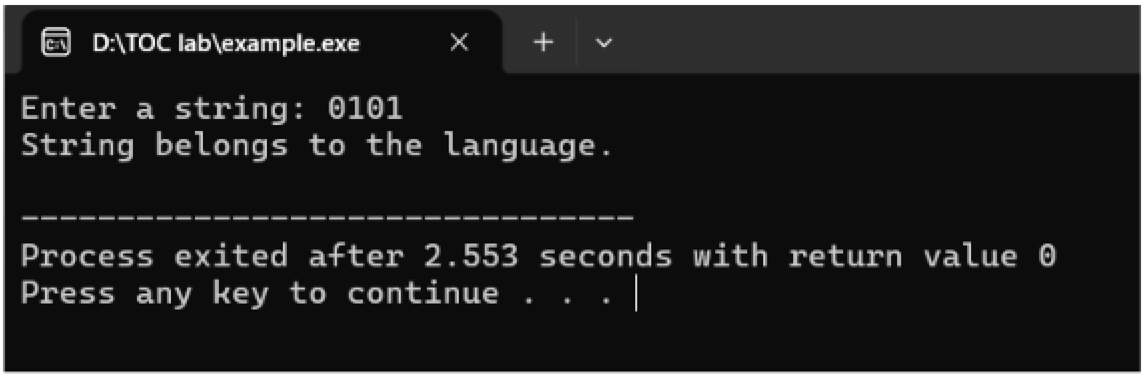
**1 Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)**

**S → 0A1 A → 0A | 1A | ε**

**C Program**

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int n;

    printf("Enter a string: ");
    scanf("%s", str);

    n = strlen(str);

    if (n >= 2 && str[0] == '0' && str[n - 1] == '1')
        printf("String belongs to the language.\n");
    else
        printf("String does not belong to the language.\n");

    return 0;
}
```

**Output:**



**Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)**

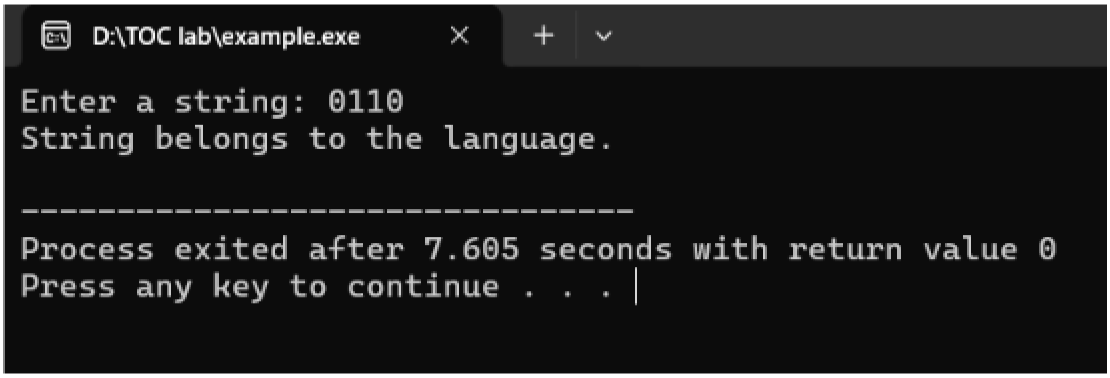
**S → 0S0 | 1S1 | 0 | 1 | ε**

**CODE:**

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int i, n, flag = 1;

    printf("Enter a string: ");
    scanf("%s", str);

    n = strlen(str);

    for(i = 0; i < n / 2; i++)
    {
        if(str[i] != str[n - i - 1])
        {
            flag = 0;
            break;
        }
    }

    if(flag == 1)
        printf("String belongs to the language.\n");
    else
```

```
        printf("String does not belong to the language.\n");

    return 0;
}
```

**Output:**




**3. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)**

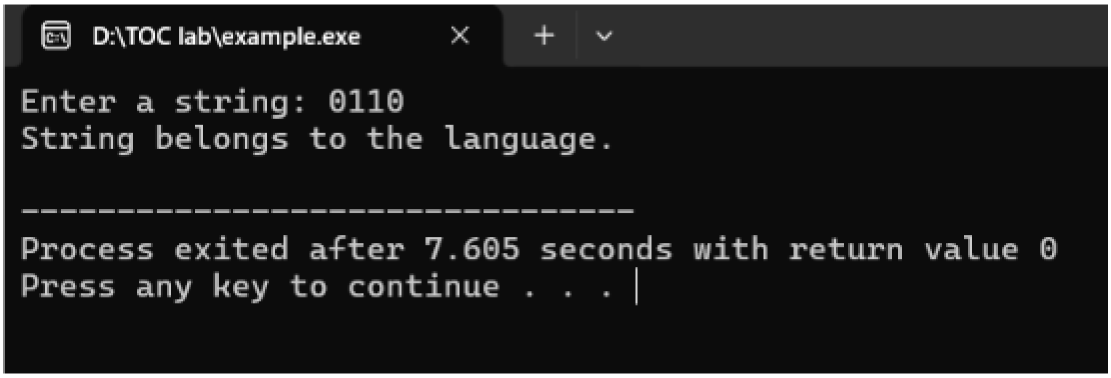
**S → 0S0 | A A → 1A | ε**

**CODE:**

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int n, i, j, valid = 1;

    printf("Enter a string: ");
    scanf("%s", str);

    n = strlen(str);

    i = 0;
```

```c
    j = n - 1;

    /* Check equal number of 0s at both ends */
    while (i < j && str[i] == '0' && str[j] == '0')
    {
        i++;
        j--;
    }

    /* Middle part must contain only 1s */
    while (i <= j)
    {
        if (str[i] != '1')
        {
            valid = 0;
            break;
        }
        i++;
    }

    if (valid)
        printf("String belongs to the language.\n");
    else
        printf("String does not belong to the language.\n");

    getchar();
    getchar();

    return 0;
}
```

**OUTPUT:**




**Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)**

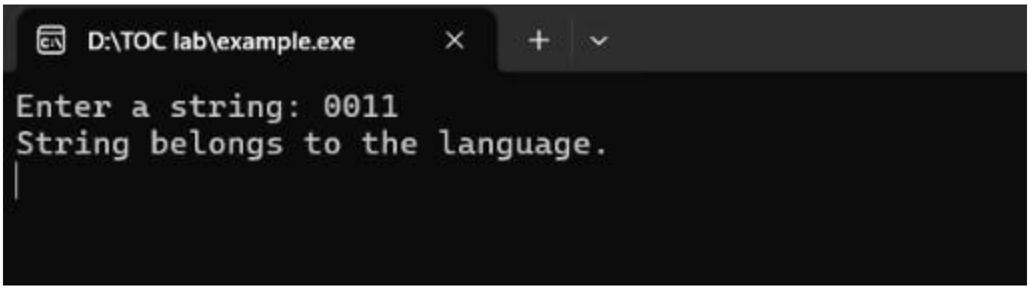
**S → 0S1 | ε**

**Code:**

```
#include <stdio.h>
#include <string.h>

int main()
{
   char str[100];
   int i, n, zeros = 0, ones = 0, valid = 1;

   printf("Enter a string: ");
   scanf("%s", str);

   n = strlen(str);

   /* Count 0s first */
   i = 0;
   while (i < n && str[i] == '0')
   {
      zeros++;
      i++;
```

```
    }

    /* Count 1s after all 0s */
    while (i < n && str[i] == '1')
    {
        ones++;
        i++;
    }

    /* Check equal number of 0s and 1s and correct order */
    if (i != n || zeros != ones)
        valid = 0;

    if (valid)
        printf("String belongs to the language.\n");
    else
        printf("String does not belong to the language.\n");

    getchar();
    getchar();

    return 0;
}
```

**OUTPUT:**



**5. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)**

**S → A101A A → 0A | 1A | ε**

**CODE:**

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int i, n, found = 0;

    printf("Enter a string: ");
    scanf("%s", str);

    n = strlen(str);

    for(i = 0; i <= n - 3; i++)
    {
        if(str[i] == '1' && str[i + 1] == '0' && str[i + 2] == '1')
        {
            found = 1;
            break;
        }
    }

    if(found)
        printf("String belongs to the language.\n");
    else
        printf("String does not belong to the language.\n");
```

```
    getchar();

    getchar();


    return 0;
}
```

**OUTPUT:**